

Mathematical Foundations for Machine Learning

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Why Mathematical Foundations?

- Machine Learning models are built on mathematical principles.
- Understanding math helps in:
 - Designing better algorithms
 - Improving model performance
 - Interpreting results
- Core areas: Linear Algebra, Calculus, and Probability.

Data Matrix Representation

Data Matrix

Data can be represented as an $n \times d$ matrix, where: n = number of instances (rows), d = number of features (columns)

$$D = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix}$$

Rows (Instances)

Rows represent data samples, records, objects, or feature vectors:

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{id})$$

Columns (Features)

Columns represent attributes, variables, dimensions, or features:

$$\mathbf{X}_j = (x_{1j}, x_{2j}, \dots, x_{nj})$$

Iris Dataset Example

Iris Dataset

- One of the most widely used datasets in Machine Learning.
- Contains 150 flower samples from three iris species:
 - Iris-setosa
 - Iris-versicolor
 - Iris-virginica
- Each sample has four features:
 - Sepal Length
 - Sepal Width
 - Petal Length
 - Petal Width

ID	X_1	X_2	X_3	X_4	Class
x_1	5.9	3.0	4.2	1.5	Versicolor
x_2	6.9	3.1	4.9	1.5	Versicolor
x_3	6.6	2.9	4.6	1.3	Versicolor
x_4	4.6	3.2	1.4	0.2	Setosa
x_5	6.0	2.2	4.0	1.0	Versicolor
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_{150}	5.1	3.4	1.5	0.2	Setosa

Iris Dataset as a Data Matrix

$$D = \begin{bmatrix} 5.9 & 3.0 & 4.2 & 1.5 \\ 6.9 & 3.1 & 4.9 & 1.5 \\ 6.6 & 2.9 & 4.6 & 1.3 \\ 4.6 & 3.2 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 5.1 & 3.4 & 1.5 & 0.2 \end{bmatrix}$$

- Rows (x_i): individual flower samples.
- Columns (X_j): measured features.
- Class labels indicate the iris species.
- Matrix size: 150×4 (excluding class label).

Types of Attributes

Attributes describe the characteristics or features of data objects.

Numeric Attributes

Values are represented by numbers.

- **Interval-scaled**

- Differences are meaningful.
- No true zero point.
- Example: Temperature ($^{\circ}\text{C}$, $^{\circ}\text{F}$)

- **Ratio-scaled**

- Differences and ratios are meaningful.
- Has a true zero point.
- Example: Age, Height, Weight

Categorical Attributes

Values belong to a set of categories.

- **Nominal**

- Only equality comparison.
- Example:

$$\text{Sex} = \{M, F\}$$

- **Ordinal**

- Equality and ordering are meaningful.
- Example: Education = $\{\text{HS, BS, MS, PhD}\}$

Data Representation: Algebraic View

Given a dataset

$$D = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix}$$

- n : number of samples
- d : number of features

Sample (stored as a row)

$$(x_{i1}, x_{i2}, \dots, x_{id})$$

Sample Vector (column form)

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{id})^T \in \mathbb{R}^d$$

Feature Vector

$$\mathbf{X}_j = (x_{1j}, x_{2j}, \dots, x_{nj})^T \in \mathbb{R}^n$$

Key Idea

- Rows represent **samples/instances**.
- Columns represent **features/attributes**.

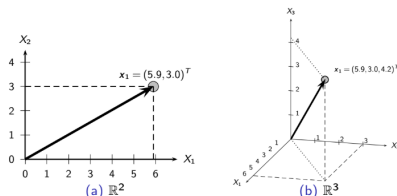
Data Representation: Geometric View

Geometric Interpretation

- Each sample is a point in a feature space.
- A dataset with d features forms a space $\mathbb{R}^d$.

Example (Iris dataset):

$$\mathbf{x} = (5.9, 3.0, 4.2, 1.5)^T$$



Each sample corresponds to a point in feature space.

Dimension vs. Visualization

- 2 features $\rightarrow$ point in $\mathbb{R}^2$
- 3 features $\rightarrow$ point in $\mathbb{R}^3$
- 4 features $\rightarrow$ point in $\mathbb{R}^4$

Vector Norm and Unit Vector

Euclidean Norm

For a vector $\mathbf{a} \in \mathbb{R}^m$,

$$\|\mathbf{a}\| = \sqrt{\mathbf{a}^T \mathbf{a}} = \sqrt{\sum_{i=1}^m a_i^2}$$

The norm measures the **length** of a vector.

Unit Vector

A vector with length 1:

$$\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|}$$

$$\|\mathbf{u}\| = 1$$

Why Important?

Many machine learning algorithms normalize data to remove the effect of scale.

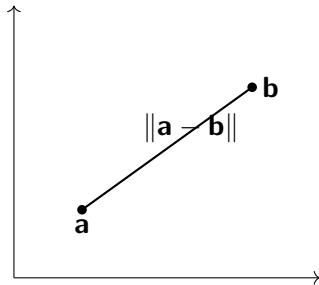
Distance Between Data Points

For two points

$$\mathbf{a}, \mathbf{b} \in \mathbb{R}^m$$

their Euclidean distance is

$$d(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\| = \sqrt{\sum_{i=1}^m (a_i - b_i)^2}$$



- Small distance $\Rightarrow$ similar points.
- Large distance $\Rightarrow$ dissimilar points.

Machine Learning Insight

Distance is the basis of k-NN, clustering, and similarity search.

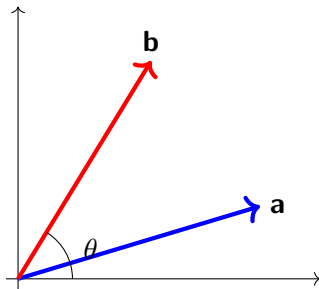
Dot Product and Angle

Dot Product

$$\mathbf{a}^T \mathbf{b} = \sum_{i=1}^m a_i b_i$$

Angle Between Vectors

$$\cos \theta = \frac{\mathbf{a}^T \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$



Interpretation

- $\theta = 0^\circ$: same direction
- $\theta = 90^\circ$: orthogonal
- $\theta = 180^\circ$: opposite direction

Similarity ↑



$\cos \theta$ ↑

Orthogonal Projection

Idea: A vector can be split into two parts:

$$\mathbf{b} = \mathbf{b}_{\parallel} + \mathbf{b}_{\perp}$$

where

- $\mathbf{b}_{\parallel}$: component parallel to $\mathbf{a}$
- $\mathbf{b}_{\perp}$: component perpendicular to $\mathbf{a}$

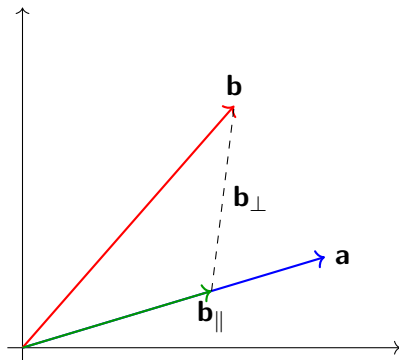
The projection of $\mathbf{b}$ onto $\mathbf{a}$ is

$$\mathbf{b}_{\parallel} = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}} \mathbf{a}$$

Key Idea

Projection gives the part of $\mathbf{b}$ that lies in the direction of $\mathbf{a}$.

Orthogonal Projection: Geometric View



$$\mathbf{b} = \mathbf{b}_{\parallel} + \mathbf{b}_{\perp}$$

Prerequisite Concepts for Eigenvalues and Eigenvectors

Before defining eigenvalues and eigenvectors, we need a few basic ideas:

- **Matrices as Linear Transformations**

- A matrix $\mathbf{A}$ transforms a vector $\mathbf{x}$ into another vector $\mathbf{Ax}$
- This transformation may rotate, stretch, or shrink the vector

- **Identity Matrix**

- Denoted by $\mathbf{I}$
- Property: $\mathbf{Ix} = \mathbf{x}$
- It acts like the number 1 in matrix algebra

- **Scalar Multiplication of Vectors**

- $\lambda \mathbf{v}$ changes only the magnitude of $\mathbf{v}$
- Direction remains unchanged

Key Idea: We will study vectors whose direction remains unchanged under matrix transformation.

Eigenvalues and Eigenvectors

Core Idea: Eigenvectors represent directions that remain unchanged when a linear transformation is applied, while eigenvalues represent how much those directions are stretched or compressed.

Mathematical Definition:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- $\mathbf{A}$: square matrix (transformation)
- $\mathbf{v}$: eigenvector (non-zero vector)
- λ : eigenvalue (scalar)

Geometric Interpretation:

- When matrix $\mathbf{A}$ acts on $\mathbf{v}$, the direction of $\mathbf{v}$ does not change
- Only its magnitude is scaled by λ
- If $\lambda > 1 \rightarrow$ stretching
- If $0 < \lambda < 1 \rightarrow$ shrinking
- If $\lambda < 0 \rightarrow$ direction flips

Eigenvalues of a 3×3 Matrix: Derivation

Given matrix:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Eigenvalue equation:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Rewriting using identity matrix:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{I}\mathbf{v}$$

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = 0$$

Condition for non-trivial solution:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

Eigenvalues of a 3×3 Matrix: Determinant Expansion

Compute:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix} = 0$$

This gives a cubic equation in λ :

$$\lambda^3 + c_2\lambda^2 + c_1\lambda + c_0 = 0$$

Steps:

- Expand determinant along any row/column
- Simplify to polynomial form
- Solve for eigenvalues λ

Eigenvalues: Clearing Common Confusion

Start from eigenvalue equation:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Correct transformation:

$$\mathbf{A}\mathbf{v} - \lambda\mathbf{v} = 0$$

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = 0$$

Key distinction:

NOT correct: $\mathbf{A} - \lambda\mathbf{I} = 0$

Correct condition: $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$

Why determinant = 0?

- We need non-zero solution $\mathbf{v} \neq 0$
- Only possible when system is singular
- Singular matrix $\Rightarrow \det(\mathbf{A} - \lambda\mathbf{I}) = 0$

Eigenvalues of a 3×3 Matrix: Example

Given matrix:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

We find eigenvalues using:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

Step 1: Form $A - \lambda I$

$$\mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 2 - \lambda & 1 & 0 \\ 1 & 2 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{bmatrix}$$

Step 2: Determinant Expansion

Expand along first row:

$$\det = (2 - \lambda) \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 0 & 2 - \lambda \end{vmatrix} + 0$$

Step-by-Step Eigenvalue Computation (3×3 Example)

Step 3: Compute Minors

$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 \quad , \quad \begin{vmatrix} 1 & 1 \\ 0 & 2-\lambda \end{vmatrix} = 2-\lambda$$

Step 4: Substitute Back

$$\begin{aligned} \det &= (2-\lambda)((2-\lambda)^2 - 1) - (2-\lambda) \\ &= (2-\lambda)((2-\lambda)^2 - 2) \end{aligned}$$

Step 5: Polynomial Form

$$\begin{aligned} (2-\lambda)^2 &= 4 - 4\lambda + \lambda^2 \\ \det &= (2-\lambda)(\lambda^2 - 4\lambda + 2) \end{aligned}$$

Step 6: Cubic Equation

$$\lambda^3 - 6\lambda^2 + 10\lambda - 4 = 0$$

Try $\lambda = 2$ (root)

Step 7: Factorization

$$(\lambda - 2)(\lambda^2 - 4\lambda + 2) = 0$$

Final Eigenvalues

$$\lambda_1 = 2, \quad \lambda_2 = 2 + \sqrt{2}, \quad \lambda_3 = 2 - \sqrt{2}$$

Eigenvector visualization

Eigenvectors: Geometric Visualization

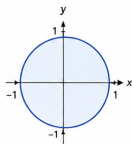
Eigenvectors are special directions that do not change direction under a linear transformation. They only get scaled.

$$A\mathbf{v} = \lambda\mathbf{v} \quad \begin{array}{l} \mathbf{v} = \text{eigenvector} \\ \lambda = \text{eigenvalue} \end{array}$$

Key Idea

- A matrix stretches/compresses space.
- In general, most directions change.
- Eigenvectors are the special directions that stay on the same line (only length changes).

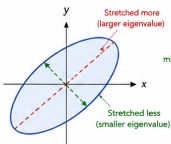
1. Original Circle



$$\text{Unit circle: } x^2 + y^2 = 1$$

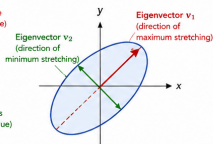
Apply linear transformation A

2. Transformed Shape (Ellipse)



A circle becomes an ellipse.

3. Eigen-directions (Axes of Ellipse)



Eigenvectors give the directions of the ellipse's axes.

Example

Consider the matrix

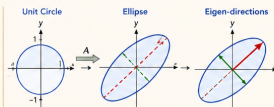
$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

Eigenvalues:

$$\lambda_1 \approx 3.618, \quad \lambda_2 \approx 1.382$$

Eigenvectors (unit):

$$\mathbf{v}_1 \approx \begin{bmatrix} 0.851 \\ 0.526 \end{bmatrix}, \quad \mathbf{v}_2 \approx \begin{bmatrix} -0.526 \\ 0.851 \end{bmatrix}$$



What Eigenvalues Do

- Eigenvalues determine how much stretching/shrinking happens in the direction of each eigenvector.

• If $\lambda > 1 \rightarrow$ stretch

If $0 < \lambda < 1 \rightarrow$ shrink

If $\lambda < 0 \rightarrow$ flip + stretch

Important Takeaways

- ✓ Circle $\rightarrow$ becomes ellipse under a linear transformation.
- ✓ Eigenvectors = directions that stay unchanged.
- ✓ Eigenvalues = amount of stretching/compression.
- ✓ In ML (PCA): eigenvectors = directions of maximum variance.

Best Correct Statement: A linear transformation maps a circle into an ellipse, where the eigenvectors determine the directions of the ellipse's axes and the eigenvalues determine the stretching along those axes.

Why Eigenvalues and Eigenvectors Matter?

Key Insight: Eigenvectors reveal the most important directions in data, and eigenvalues measure their strength.

Interpretation in Data Analysis:

- Eigenvectors = principal directions of variance in data
- Eigenvalues = amount of variance captured along each direction

In Machine Learning:

- Used in PCA (Principal Component Analysis)
- Helps reduce dimensionality while preserving structure
- Identifies most informative features

Intuition:

- Think of data as a cloud of points
- Eigenvectors show the orientation of the cloud
- Eigenvalues show how stretched the cloud is in each direction

Differentiation: Idea and Intuition

Concept: Differentiation measures how a function changes when its input changes.

Mathematical Definition:

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- Represents the **slope of the tangent line**.
- Measures the rate of change of a function.
- Indicates how sensitive the output is to small input changes.

Interpretation:

- $\frac{df}{dx} > 0$: Function is increasing
- $\frac{df}{dx} < 0$: Function is decreasing
- $\frac{df}{dx} = 0$: Potential minimum or maximum

Why is differentiation important in Machine Learning?

Training a model means finding parameters that minimize a loss function:

$$L(\theta)$$

Key Idea:

$$\frac{dL(\theta)}{d\theta} = 0$$

helps identify critical points where the loss may be minimum or maximum.

Example:

$$f(x) = x^2$$

$$\frac{df(x)}{dx} = 2x$$

At $x = 0$:

$$\frac{df(x)}{dx} = 0$$

which corresponds to the minimum value of the function.

Gradient: Extension to Multiple Variables

Problem:

Most Machine Learning models depend on many parameters rather than a single variable.

For a multivariable function

$$f(x_1, x_2, \dots, x_n)$$

the gradient is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

- Collection of all partial derivatives.
- Shows how the function changes along each dimension.
- Generalizes differentiation to higher dimensions.

Gradient Descent: Why Move Opposite to the Gradient?

Geometric Interpretation:

- Gradient points toward the direction of **steepest increase**.
- Its magnitude indicates how steep the increase is.

To minimize a loss function, we move in the opposite direction.

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

where

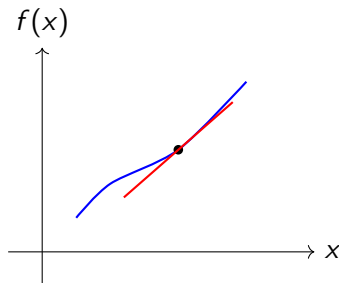
- θ = model parameters
- η = learning rate
- $\nabla L(\theta)$ = gradient of the loss function

Mountain Analogy:

- Gradient points uphill.
- Opposite gradient points downhill.
- Gradient Descent follows the downhill path to reach the minimum loss.

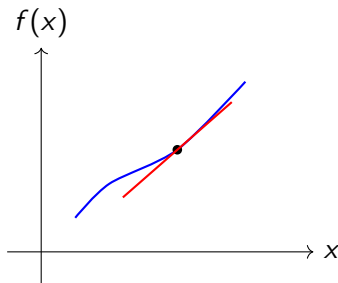
Differentiation: Intuition (Visual View)

Idea: Differentiation measures the slope of a curve at a point.



Differentiation: Intuition (Visual View)

Idea: Differentiation measures the slope of a curve at a point.



- The red line shows the **tangent (instantaneous slope)**
- Slope changes at every point of the function

Differentiation: Step-by-Step Understanding

- Step 1: Choose a function $f(x)$
- Step 2: Slightly change input $x \rightarrow x + h$
- Step 3: Measure output change

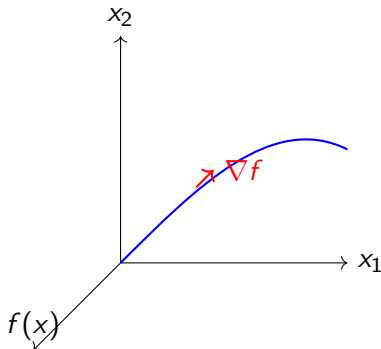
$$\frac{f(x + h) - f(x)}{h}$$

- Step 4: Let $h \rightarrow 0 \rightarrow$ exact slope (derivative)

Result: Instant rate of change at a point.

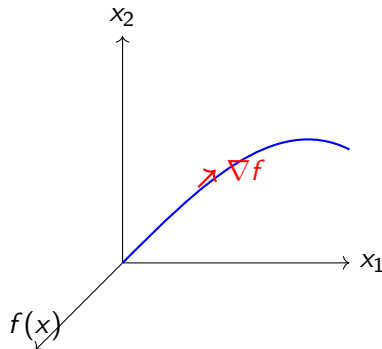
Gradient: Geometric Intuition

A gradient shows direction of fastest increase in a surface.



Gradient: Geometric Intuition

A gradient shows direction of fastest increase in a surface.



- Gradient points uphill (steepest ascent)
- Negative gradient $\rightarrow$ direction of descent

Gradient Descent: Learning Process

Goal: Minimize loss function $L(x)$

Update Rule:

$$x_{t+1} = x_t - \eta \nabla L(x_t)$$

- Start with random parameters
- Compute gradient (direction of increase)
- Move in opposite direction
- Repeat until convergence

Intuition: Walking downhill to reach the lowest point.

ML Connection Summary

- Differentiation $\rightarrow$ slope in 1D
- Gradient $\rightarrow$ slope in multiple dimensions
- Gradient Descent $\rightarrow$ optimization engine of ML

Key Insight: All modern machine learning models are trained by minimizing loss using derivatives.

Why Do We Need Distributions?

Intuition: A distribution describes how data values are spread and how frequently they occur.

Example: Student Heights

- Most students have heights around 165–175 cm.
- Very short and very tall students are less common.
- The pattern of occurrence of heights forms a **distribution**.

A distribution answers:

- Which values are common?
- Which values are rare?
- How likely is a value to occur?

"A distribution is a mathematical description of how data is spread." A distribution in Machine Learning is a mathematical description of how data, labels, errors, features, or predictions are spread and how likely different values are to occur. Most ML algorithms either assume a distribution, estimate a distribution, or predict a distribution.

Data: Probabilistic View

Random Variable

A random variable is a function

$$X : \Omega \rightarrow \mathbb{R}$$

that maps each outcome in the sample space Ω to a real number.

Types of Random Variables

- **Discrete:** takes a finite or countable set of values.
- **Continuous:** takes any value within an interval of $\mathbb{R}$.

For a numeric attribute X_j , we often use the identity mapping:

$$X(v) = v, \quad v \in \mathbb{R}$$

Interpretation

In machine learning, each feature can be viewed as a random variable describing the data-generating process.

Example: Discrete Random Variable

Consider the Iris feature **Sepal Length**.

Define a binary random variable A :

$$A(v) = \begin{cases} 0, & v < 7 \\ 1, & v \geq 7 \end{cases}$$

Interpretation

- $A = 0$: Sepal length is less than 7 cm.
- $A = 1$: Sepal length is at least 7 cm.

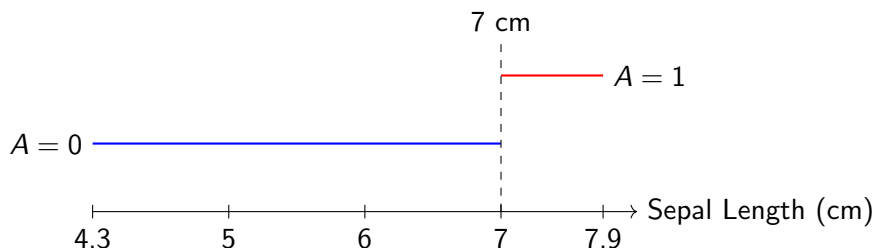
For the Iris dataset:

$$\Omega = [4.3, 7.9]$$

$$\text{Range}(A) = \{0, 1\}$$

Since A can take only two values, it is a **discrete random variable**.

Discrete Variable Example



$$A(v) = \begin{cases} 0, & v < 7 \\ 1, & v \geq 7 \end{cases}$$

Probability Mass Function (PMF)

Definition

For a discrete random variable X , the probability mass function (PMF) is

$$f(x) = P(X = x), \quad x \in \mathbb{R}$$

Properties of a PMF

- 1 Non-negativity:

$$f(x) \geq 0$$

- 2 Total probability equals 1:

$$\sum_x f(x) = 1$$

A discrete random variable assigns probability only to specific values. The probability is concentrated (or *massed*) at those values and is zero elsewhere.

Probability Mass Function (PMF)

Definition

$$f(x) = P(X = x)$$

Properties

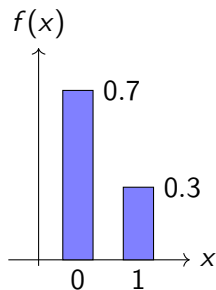
$$f(x) \geq 0$$

$$\sum_x f(x) = 1$$

Example

Suppose

$$X \in \{0, 1\}$$



Sepal Length: Bernoulli Distribution

Iris Dataset Example

Define a binary random variable A :

$$A(v) = \begin{cases} 0, & v < 7 \\ 1, & v \geq 7 \end{cases}$$

- $A = 1$: Iris has a **long sepal** (≥ 7 cm)
- $A = 0$: Otherwise

From the 150 Iris samples:

$$\#(A = 1) = 13, \quad \#(A = 0) = 137$$

Therefore,

$$P(A = 1) = \frac{13}{150} = 0.087 = p$$

$$P(A = 0) = \frac{137}{150} = 0.913 = 1 - p$$

Bernoulli Probability Mass Function

A Bernoulli random variable takes only two values:

$$A \in \{0, 1\}$$

Its PMF is

$$f(a) = \begin{cases} p, & a = 1 \\ 1 - p, & a = 0 \end{cases}$$

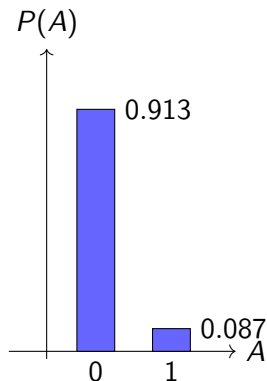
For the Iris dataset:

$$p = P(A = 1) = 0.087$$

$$1 - p = P(A = 0) = 0.913$$

A randomly selected Iris flower has only an 8.7% chance of having a sepal length of at least 7 cm.

Bernoulli Distribution for Long Sepal Length



$$P(A = 0) = 0.913, \quad P(A = 1) = 0.087$$

Binomial Distribution

Question:

What is the probability of finding k Irises with long sepals (≥ 7 cm) in m independent trials?

Let

$$p = P(\text{Long Sepal}) = 0.087$$

and

B = Number of successes in m trials

Then

$$B \sim \text{Binomial}(m, p)$$

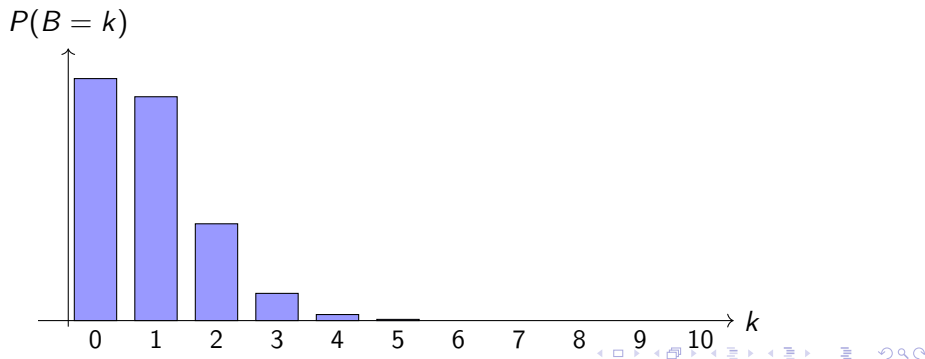
$$P(B = k) = \binom{m}{k} p^k (1 - p)^{m-k}$$

Binomial Distribution Example

For Iris data:

$$p = 0.087, \quad m = 10$$

$$P(B = k) = \binom{10}{k} (0.087)^k (0.913)^{10-k}$$



Probability Density Function (PDF)

Continuous Random Variable

For a continuous variable X ,

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Properties:

$$f(x) \geq 0$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Key Difference

Discrete: Probability is assigned to points. Continuous: Probability is assigned to intervals (areas under the curve).

Normal Distribution (Gaussian Distribution)

Many real-world continuous variables such as height, weight, and flower measurements, can often be approximated by a normal distribution.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Example: Iris Sepal Length

$$\mu = 5.84, \quad \sigma^2 = 0.681$$

- **Mean** (μ) = 5.84 cm represents the center of the data.
- **Variance** (σ^2) = 0.681 measures the spread of the data.
- **Standard Deviation** (σ) = $\sqrt{0.681} = 0.825$ cm.
- The distribution is symmetric around the mean.
- Values closer to μ occur more frequently than values far from μ .

Interpreting the Normal Distribution

For Iris sepal length:

$$\mu = 5.84, \quad \sigma = 0.825$$

- About 68% of samples lie within:

$$\mu \pm \sigma = [5.02, 6.67]$$

- About 95% of samples lie within:

$$\mu \pm 2\sigma = [4.19, 7.49]$$

- About 99.7% of samples lie within:

$$\mu \pm 3\sigma = [3.37, 8.32]$$

Cumulative Distribution Function

The CDF gives the probability that a variable is less than or equal to a value.

$$F(x) = P(X \leq x)$$

For discrete variables:

$$F(x) = \sum_{u \leq x} f(u)$$

For continuous variables:

$$F(x) = \int_{-\infty}^x f(u) du$$

Interpretation

CDF accumulates probability from left to right.

Joint Probability Distribution

Consider two binary variables:

$$X_1 = \begin{cases} 1, & \text{Long Sepal Length} \\ 0, & \text{Otherwise} \end{cases}$$

$$X_2 = \begin{cases} 1, & \text{Long Sepal Width} \\ 0, & \text{Otherwise} \end{cases}$$

The joint PMF is

$$f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$$

It measures the probability of observing both events simultaneously.

Random Samples and Independence

A random sample consists of

$$S_1, S_2, \dots, S_n$$

drawn independently from the same distribution.

This is called:

IID

- Independent
- Identically Distributed

Two variables are independent if

$$P(X_1, X_2) = P(X_1) P(X_2)$$

Why Important?

Many machine learning models assume IID data.

Sample Statistics

Statistics summarize data using a sample.

Sample Mean

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

For Iris sepal length:

$$\hat{\mu} = 5.84$$

Sample Variance

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

$$\hat{\sigma}^2 = 0.681$$

Mean measures central tendency, while variance measures spread.

Probability Distributions

- Describe uncertainty in data.
- **Types:**
 - Gaussian (Normal): Bell-shaped distribution
 - Bernoulli: Binary outcomes (0 or 1)
 - Binomial: Multiple Bernoulli trials
- Key idea: Assign probabilities to events.

Definition: Average value of a random variable.

Discrete:

$$\mathbb{E}[X] = \sum xP(x)$$

Continuous:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x) dx$$

- Represents the long-term average outcome.
- Used in decision making and risk analysis.

Expectation Example

Uniform Distribution on $[0, 1]$

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} \mathbb{E}[X] &= \int_0^1 xf(x) dx = \int_0^1 x dx \\ &= \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2} \end{aligned}$$

Interpretation

All values between 0 and 1 are equally likely, so the average value is

$$\mathbb{E}[X] = 0.5.$$

Variance

Definition: Variance measures the spread of a random variable around its mean.

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2], \quad \mu = \mathbb{E}[X]$$

Equivalent form:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

- Measures how far values typically lie from the mean.
- High variance $\rightarrow$ data is widely spread.
- Low variance $\rightarrow$ data is tightly clustered.
- Always non-negative.

Interpretation

Variance is the average squared deviation from the mean. It quantifies uncertainty in the data.

Let X be the outcome of a fair die:

$$X \in \{1, 2, 3, 4, 5, 6\}, \quad P(X = x) = \frac{1}{6}$$

Step 1: Expectation

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

Step 2: Compute $\mathbb{E}[X^2] = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6}$

Step 3: Variance

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\text{Var}(X) = \frac{91}{6} - (3.5)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

Interpretation

Even though the mean is 3.5, outcomes spread around it. Variance measures this spread.

Bayes' Theorem

Idea: Bayes' Theorem helps us update our belief about an event when new evidence is observed.

Formula:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Interpretation of terms:

- $P(A|B)$: **Posterior** → Updated probability after observing evidence B
- $P(B|A)$: **Likelihood** → Probability of observing evidence B if A is true
- $P(A)$: **Prior** → Initial belief before seeing evidence
- $P(B)$: **Evidence** → Overall probability of observing B

Key Insight

Bayes' Theorem reverses conditional probability: it allows us to infer causes from observed effects.

Bayes' Theorem Example (Medical Test)

Problem Setup:

- Disease prevalence: $P(D) = 0.01$
- Test sensitivity: $P(T|D) = 0.99$
- False positive rate: $P(T|\neg D) = 0.05$

Question: If a person tests positive, what is $P(D|T)$?

Step 1: Total probability of positive test

$$P(T) = P(T|D)P(D) + P(T|\neg D)P(\neg D)$$

$$P(T) = (0.99)(0.01) + (0.05)(0.99) = 0.0099 + 0.0495 = 0.0594$$

Step 2: Apply Bayes' Theorem

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)} = \frac{0.0099}{0.0594} \approx 0.1667$$

Insight Even with a positive test, the probability of having the disease is only about **16.7%** due to low prevalence.

Applications: Spam detection, medical diagnosis, classification.

Summary

- Linear Algebra $\rightarrow$ Data representation and transformations
- Calculus $\rightarrow$ Optimization and learning
- Probability $\rightarrow$ Uncertainty and prediction
- These form the backbone of Machine Learning

Thank You